Project Report

Samsung Supply Chain & Logistics Analytics Dashboard using Power BI

1. Project Title

Samsung Supply Chain & Logistics Analytics Dashboard using Power BI

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2. Introduction

Efficient supply chain management is one of the most critical aspects of modern businesses. Companies like Samsung operate across multiple suppliers, manufacturing plants, warehouses, logistics partners, and sales platforms, generating vast amounts of operational data every day. Analyzing this data manually is time-consuming and often results in delayed decision-making.

This project focuses on developing an interactive **Supply Chain & Logistics Analytics Dashboard** using **Power BI** to provide real-time visibility into the entire supply chain process. The dashboard enables stakeholders to monitor operational performance, evaluate supplier efficiency, optimize inventory levels, analyze logistics operations, and track overall business KPIs. The solution transforms raw supply chain data into meaningful insights that support faster and data-driven decision-making.

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3. Problem Statement

Samsung manages a complex supply chain network involving procurement, manufacturing, warehouse management, transportation, and customer delivery. The large volume of operational data makes it difficult to identify inventory shortages, supplier delays, shipment issues, and inefficient logistics processes.

The primary objective of this project is to build an interactive Power BI dashboard that consolidates supply chain data from multiple sources, monitors key business metrics, identifies operational bottlenecks, and provides actionable insights to improve inventory management, logistics efficiency, supplier performance, and overall supply chain operations.

4. Objectives

The major objectives of the project are:

- Develop an end-to-end Supply Chain Analytics Dashboard using Power BI.
- Clean and transform raw datasets using Power Query.
- Design an optimized Star Schema data model.
- Calculate business KPIs using DAX.
- Monitor inventory, logistics, suppliers, and warehouse performance.
- Track shipment delays and delivery success rates.
- Compare revenue across different sales channels.
- Support strategic business decisions using interactive dashboards.

5. Dataset Description

The project uses multiple datasets representing different components of Samsung's supply chain ecosystem.

The datasets include:

- Product Details
- Supplier Information
- Sales Data
- Inventory Records
- Warehouse Details
- Shipment Information
- Customer Orders
- Date Dimension

These datasets collectively represent the complete flow of products from procurement to final customer delivery.

6. Methodology

6.1 Data Ingestion

The project follows a dynamic and scalable data loading approach.

- Imported multiple datasets using **Power Query**.
- Implemented **Folder-Based Import** to automatically load data files.
- Created a **Parameterized File Path** to ensure the dashboard can be easily shared across different systems without modifying file locations.
- Standardized data formats before loading into the model.

6.2 Data Cleaning & Transformation

Power Query was extensively used for ETL (Extract, Transform, Load) operations.

The following transformations were performed:

- Removed duplicate records
- Handled missing values
- Renamed columns
- Changed data types
- Split and merged columns where required
- Standardized categorical values
- Removed unnecessary fields
- Created calculated columns
- Prepared clean datasets for analysis

These transformations improved data quality and reporting accuracy.

6.3 Data Modeling

A **Star Schema** architecture was implemented to improve reporting performance and simplify analysis.

Fact Tables

- Sales
- Inventory
- Shipments

Dimension Tables

- Product
- Supplier
- Warehouse
- Customer
- Date

Relationships were established using **One-to-Many** cardinality to enable efficient filtering and accurate calculations throughout the dashboard.

The optimized data model reduced redundancy and improved dashboard performance.

6.4 DAX Measures & KPI Development

A dedicated **Measure Table** was created to organize all business calculations.

Several DAX measures were developed to monitor operational and financial performance.

Business KPIs

- Total Revenue
- Total Profit
- Profit Margin
- Total Orders
- Total Shipments

Supply Chain KPIs

- Inventory Turnover
- Perfect Order Percentage
- Average Lead Time
- Delayed Shipments
- Defective Units
- Safety Stock Analysis
- Warehouse Utilization

Complex DAX formulas were also used to troubleshoot missing delivery statuses and improve calculation accuracy.

6.5 Dashboard Design

The dashboard was designed with a business-focused layout that enables users to analyze the complete supply chain process.

Major design features include:

- Executive KPI Cards
- Supply Chain Flow Visualization
- Inventory Dashboard
- Supplier Performance Dashboard
- Warehouse Dashboard
- Logistics Dashboard
- Sales Channel Analysis
- Interactive Slicers
- Drill-through Pages
- Dynamic Tooltips
- Geographic Maps
- Theme Automation using Power BI Studio

The dashboard allows users to explore data interactively across different business dimensions.

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7. Key Performance Indicators (KPIs)

The dashboard monitors several important business metrics:

- Total Revenue
- Total Profit
- Profit Margin
- Inventory Turnover
- Average Lead Time
- Perfect Order Percentage
- Shipment Delay Rate
- Defect Rate
- Safety Stock
- Warehouse Utilization
- Platform-wise Revenue
- Supplier Performance Score

These KPIs help management evaluate operational efficiency and make informed business decisions.

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8. Business Insights

The dashboard provides valuable insights such as:

- Identification of suppliers with consistently high lead times.
- Detection of delayed shipments affecting customer satisfaction.
- Analysis of inventory levels to minimize stock-outs and overstocking.
- Comparison of sales generated through Amazon and Direct Store channels.
- Evaluation of warehouse efficiency and stock movement.
- Monitoring of product defects and delivery performance.
- Identification of logistics bottlenecks impacting overall operations.

These insights enable management to optimize supply chain performance and reduce operational costs.

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9. Tools & Technologies Used

Tool	Purpose
Power BI	Dashboard Development
Power Query	ETL & Data Cleaning
DAX	Business Calculations
Data Modeling	Star Schema Design
Excel/CSV	Data Source
Power BI Studio	Theme & UI Design

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10. Challenges Faced

During the project, several challenges were encountered:

- Managing relationships across multiple datasets.
- Handling missing and inconsistent data.
- Designing an optimized Star Schema.

- Writing complex DAX measures for business KPIs.
- Debugging incorrect filter contexts.
- Maintaining dashboard performance with multiple visuals.
- Creating a scalable data import process.

These challenges were addressed through proper data modeling, Power Query transformations, and optimized DAX calculations.

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11. Business Impact

The developed dashboard helps Samsung:

- Improve inventory planning.
- Reduce stock-outs and excess inventory.
- Monitor supplier reliability.
- Improve delivery performance.
- Optimize warehouse operations.
- Increase operational visibility.
- Support faster, data-driven decision-making.
- Enhance overall supply chain efficiency.

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12. Conclusion

The **Samsung Supply Chain & Logistics Analytics Dashboard** successfully demonstrates an end-to-end Business Intelligence solution using Power BI. The project covers the complete analytics workflow, including data ingestion, ETL, data modeling, DAX development, and interactive dashboard design. By integrating multiple supply chain datasets into a unified analytical platform, the dashboard enables stakeholders to monitor key performance indicators, identify operational bottlenecks, and make informed business decisions.

This project showcases practical skills in **Power BI, Power Query, DAX, data modeling, ETL, KPI development, and business intelligence**, making it a strong portfolio project for Data Analyst and Business Intelligence roles. It reflects the ability to convert raw operational data into meaningful insights that drive business value and improve organizational efficiency.